| NOT MEASUREMENT SENSITIVE |

MIL-B-29159A(YD)
13 January 1989
SUPERSEDING
MIL-B-29159(YD)
23 February 1976

MILITARY SPECIFICATION

BOAT SYSTEM, UTILITY, NOMINAL, 20-FOOT REINFORCED PLASTIC WITH 100-HORSEPOWER OUTBOARD ENGINE AND TRAILER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size reinforced plastic commercial utility boat system, with 100-horsepower (hp), outboard engine trailer, and all other necessary components and accessories to provide a complete, safe, usable boat for use in placing and handling oil spill barriers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock.
PPP-P-40 - Preservation and Packing of Hand Tool; Tool and Tool Accessories for Power Driven, Metal and Woodworking Machinery.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping Fiberboard.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the document cited in the solicitation (see 6.2).

U.S. COAST GUARD REGULATIONS (USCO)

USCO - Title 46, Code of Federal Regulation, Parts 24 and 25.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

AMERICAN BOAT AND YACHT COUNCIL (ABYC)

ABYC - Safety Standards for Small Craft.

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:47Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Boat and Yacht Council, Inc., 15 East 26th Street, Room 1603, New York, NY 10010.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM A 473 - Stainless and Heat-Resisting Steel Forgings.
ASTM A 743 - Castings, Iron-Chromium, Iron-Chromium-Nickel and Nickel-Base Corrosion-Resistant for General Application.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC., (AWS)

AWS D1.1 - Structural Welding Code, Steel.
AWS D1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, Miami, FL 33135.)

NATIONAL MARINE MANUFACTURERS ASSOCIATION (NMMA)

NMMA Certification Handbook
Trailer Manufacturer's Association (TMA) Handbook

(Application for copies should be addressed to the National Marine Marine Manufacturers Association, 401 North Michigan Avenue, Suite 1150, Chicago, IL 60611.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

TIRE AND RIM ASSOCIATION, INC. (T&RA)

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:47Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The utility boat system shall consist of a manufacturer's standard, NMMA certified, nominal 20-foot, reinforced plastic boat with trailer and 100-hp outboard engine. The boat shall be a multiple V-hull type with chines, open cockpit, and forecastle. All accessories and safety equipment required for safe, efficient operation shall be furnished as specified herein. The engine shall be a commercial, two-cycle, long shaft, electric starting, outboard type. The trailer shall be a commercial boat trailer having a minimum rated capacity of 3,000 pounds (lb). The trailer shall be certified by the Trailer Manufacturer's Association (TMA).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Reinforced plastic hull material.

3.3.1.1 Reinforcing material. Reinforcing material shall be either fiberglass cloth or a suitable synthetic fiber substitute. Synthetic fiber laminate shall have strength characteristics equivalent or superior to the fiberglass laminate it replaces.

3.3.1.2 Polyester resin. Polyester resin shall be the manufacturer's standard fire retardant type, for use with the reinforcing material and the hull buoyancy material used.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T21:47Z
Check the source to verify that this is the current version before use.

3.3.2 Hull buoyancy material. Buoyancy material shall be of suitable density and in sufficient quantity to support the flooded boat, plus one battery weighing 40 lb, fully loaded fuel tanks, one outboard engine weighing 400 lb, and an additional 1,000 lb. The buoyancy material shall be sealed in watertight compartments or sandwiched between hull walls. The boat shall have a freeboard of not less than 1 inch.

3.3.3 Metal. Metal used in the utility boat system shall have sufficient yield strength and section modulus to withstand imposed loadings. Additionally, all exposed metal surfaces shall be provided with protection from a saltwater corrosive environment (see 3.12).

3.4 Standard commercial product. The utility boat system shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product shall be included in the utility boat system being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs, or brochures and represents the latest production model.

3.5 Boat design. The boat shall be of reinforced plastic construction. The boat shall have a transom compatible with a long shaft outboard engine. The boat shall be equipped with a self-draining outboard mounting well. The well shall have provisions for installation of a shroud, if required (see 6.2), to prevent excessive water entry. The boat shall have a fitting for drainage of accumulated water, either on land or when cruising. The boat shall be rated for an outboard engine capacity of no less than 135 hp. Construction shall be strong and durable, suitable for salt water use, and shall conform to ABYC, USCG, and NMMA standards. The boat shall have good seakeeping qualities for a boat of this size at speeds up to 30 knots.

3.5.1 Reinforced plastic construction. Reinforced plastic construction shall consist of an outer and inner shell of fiberglass or suitable synthetic fiber reinforced polyester resin with buoyancy material between the two shells. The outer shell shall be a minimum of 3/16-inch thick at the chine and 1/4-inch thick at the keel. Reinforcing material shall contribute adequate strength to the structure to resist static and dynamic loads.

3.5.2 Structural strength. The boat shall have sufficient structural reinforcement to support the engine during all required operational tests without damage to the hull or bracing and without loosening or deformation of mounting fixtures and bolts.

3.5.3 Overall boat specification.

Length	18 to 21 ft
Beam	6-1/2 to 8 ft
Freeboard with 1500-lb load	
Amidship	16 +0/-2 inches

Source: https://assist.dla.mil -- Downloaded: 202640-20T21:47Z
Check the source to verify that this is the current version before use.

Transom height	Nominal 20 inches suitable for longshaft outboard
Weight	Not to exceed 1,900 lb
Finish (exterior and interior)	Impact resistant impregnated gel coat

3.5.4 Interior configuration.

3.5.4.1 Console. The steering/engine control console shall be positioned 1/3 to 1/2 of the length of the boat behind the bow. The console shall be constructed of material similar to the hull material. The console size shall allow for ease of personnel passage between the bow and stern of the boat. Unless otherwise specified (see 6.2), the console shall be positioned on the starboard side of the boat. The console shall be equipped with a rigid windscreen of console width. The windscreen shall be a minimum of 18 inches high and shall be provided with adequate framing and bracing to ensure safe operation. Windscreen material shall meet ABYC recommendations. All gauges and engine controls shall be mounted on the console.

3.5.4.1.1 Steering. A 15-inch (minimum) diameter steering wheel operating a rack and pinion steering with a push-pull cable to the transom shall be provided. The cable shall be fastened along the interior of the boat side. Steering brackets appropriate to the supplied outboard engine shall be furnished to connect the engine to the steering cable.

3.5.4.1.2 Tachometer. An appropriate (8,000 revolutions per minutes (rpm)) heavy-duty tachometer and connections shall be installed to monitor engine speed and prevent excessive engine loading. The tachometer shall be provided with a red line at the maximum engine rated rpm.

3.5.4.1.3 Cooling water. When specified (see 6.2), an appropriate water temperature or water pressure gauge and connections shall be installed to monitor cooling water when the engine is used under heavy loading.

3.5.4.1.4 Ammeter. When specified (see 6.2), an appropriate ammeter gauge and connections shall be installed to monitor the electrical charging system.

3.5.4.1.5 Engine log meter. When specified (see 6.2), an appropriate engine operating log meter and connections shall be installed which record operation usage time in hours. An engine log meter having a totalizing mechanism of not less than 9,000 hours shall be furnished. The meter shall accurately record the total engine operating time and provide trouble-free performance under the operating conditions indicated in this specification.

3.5.4.2 Seat. One swivel-type contoured seat made of fiberglass reinforced plastic shall be positioned on a cast marine aluminum pedestal and bolted in place, directly astern of the console. The bucket seat shall be cushioned and covered with waterproof material and shall have no arms. The seat is to be positioned so as to allow operator to both stand and sit behind the steering wheel.

Source: https://assist.dla.mil -- Downloaded: 20260-20T21:47Z
Check the source to verify that this is the current version before use.

3.5.4.3 Stern section. The stern section shall be left undecked so as to allow ventilation of, and access to, portable type engine fuel tanks where utilized, electrical connections, and battery box.

3.5.4.4 Railing. Approved marine railing shall be installed on the boat. The side railing shall be manufacturer's standard length 4-inch railing and shall be installed on both sides of the boat. The bow railings shall be the manufacturer's standard bow railing.

3.5.4.5 Floor covering. The floor of the interior shall be equipped with a medium grit nonskid surface with a high contrast color.

3.5.5 Boat cover. A full length, form fitted, easily installed cover of cotton duck shall be furnished with each boat to provide protection during its storage.

3.5.6 Towing accessories. The boat shall be equipped with a towing device located at the turning center of the boat. The towing device shall be designed, as a minimum, to withstand six times the rated thrust of the boat engine-propeller combination without failure, deformation or cracking of the towing device or the boat. The design of the towing device shall also prevent interference between the engine and the article being towed.

3.5.7 Mooring attachments. The boat shall be equipped with at least five mooring cleats, one located in each corner and one at the prow. A bow eye, used for boat mooring and towing shall be installed on the forward side of the stem.

3.5.8 Hoisting fittings. The boat shall have three hoisting fittings inside the boat, two of which shall be located at the transom, port and starboard, and one located forward in the stem. Each hoist fitting, and its mounting, shall be rated for loads through 2,000 lb.

3.5.9 Safety equipment. Each utility boat system shall be equipped with USCG approved safety equipment required for a boat of the size and type specified herein. The equipment shall be mounted along the boat inside walls using appropriate mounting brackets. All equipment shall satisfy subchapter C, 46 Code of Federal Regulation, parts 24 and 25. Equipment shall include, but not be limited to the following.

2 each - 5-foot long wooden paddles.
1 each - 8-foot aluminum boat hook.
1 each - 5-lb dry chemical Purple-K-Powder fire extinguisher.
1 each - First aid kit.
1 each - Emergency flare kit.
1 each - 15-lb. Danforth anchor.
1 each - 100-foot length of 1/2-inch nylon mooring line with stainless steel thimble and eye splice at one end.
6 each - Life preservers, jacket type, fibrous glass filled.
4 each - Fastened in place rubber foot step pads.
1 each - Compressed air fog horn.
2 each - Vinyl, foam-filled, 24-inch-long fenders.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

3.5.10 Electrical accessories. The boat shall be furnished with running lights conforming to USCG regulations and requirements (subchapter C 46 CFR, parts 24 and 25) for International Waters, complete with wiring for 12 volts, direct current (dc), and waterproof switch-mounted on the control console. One variable direction, portable, switch-operated, 12-volt, sealed beam spotlight of not less than 35,000 candlepower shall be installed on the control console. The spotlight shall have a 6-foot stretch cord with waterproof plug connected into a waterproof outlet. A heavy duty marine type battery designed for starting the furnished engine shall be provided in a dry charged condition, in a plastic battery box with cover. The battery box shall be secured to the boat floor with a removable strap in a convenient location in the stern of the boat.

3.5.11 Rubber bumpers. A rubber D-type bumper shall be installed at the leading vertical edge of the hull. This fendering system shall be installed on the full perimeter of the boat except for the transom area. An additional bumper shall be installed at the waterline, on the side of the hull, if this is not the leading vertical edge. The D-type fender material shall extend not less than 3 inches from the hull and shall have a minimum vertical dimension of not less than 3 inches when installed.

3.6 Engine design. The engine shall be a commercial grade, two-cycle, long shaft, electric starting, water-cooled outboard engine rated at 100 hp minimum, but not to exceed the boat's maximum rated engine power. The engine shall be installed with all attachments and connections assembled and functioning. The motor leg shall be streamlined to minimize drag and prevent water from entering the boat over the transom. In addition, the required engine shall have the following feature which deviates from a standard recreational outboard motor: an engine speed limiting ignition to prevent engine overspeed damage when operating with light load or when propeller comes out of the water.

3.6.1 Performance. Under all test conditions specified herein, the outboard motor shall start, run, and stop without loosening, failure, or permanent deformation or excessive wear of any components in ambient temperatures not less than 32 degrees Fahrenheit (^{o}F) and not greater than 120°F. When the engine propeller combination is operated at the speed corresponding to the manufacturer's continuous duty rating, the static thrust shall be not less than 800 lb. The supplier shall submit to the contracting officer a maximum recommended operating speed in rpm for the outboard motor. If cavitation is experienced during the tests specified herein, the engine speed shall either drop off or the engine shall be able to sustain the overspeed without failure.

3.6.2 Controls. A single, manually operated control lever, completely mechanical, which regulates engine throttling, gear shifting and provides a receptacle for the ignition key, shall be provided. The control mechanism shall be equipped with appropriate length control cables and connectors, fastened along the interior side of the boat, to link the mechanism to the furnished outboard engine. The control lever shall be capable of operating the engine through a complete engine cycle (neutral to maximum forward, to neutral, to maximum reverse, to neutral) repeatedly without loosening,

Source: https://assist.dla.mil -- Downloaded: 20260-20T21:47Z
Check the source to verify that this is the current version before use.

permanent deformation or failure of any component. All controls shall be designed for operation by personnel wearing cold weather gloves. All controls and their positions shall be clearly identified (see 3.15.3).

3.6.3 Choke. A remote control manually operated choke switch with connections to the engine shall be provided.

3.6.4 Fuel system. A means for manual priming of the fuel system shall be furnished.

3.6.4.1 Carburetor. The carburetor(s) shall be the Venturi type with fixed high and low speed jets.

3.6.4.2 Fuel tanks. Manufacturer's standard fuel tanks shall be provided. Tanks and tank installation shall meet ABYC, USCG and NMMA standards. Minimum overall fuel capacity shall be 20 gallons. Fuel level indication in the tank or tanks shall be provided. If multiple tanks are utilized, they shall be valved to allow concurrent or individual feeding. Filler caps shall be attached to the tanks by detachable chain. If portable fuel tanks are utilized, they shall be securely positioned in place, utilizing removable type floor harness straps along the transom of the boat.

3.6.4.2.1 Fuel lines. Fuel lines shall be installed to prevent chafing by moving parts or vibration, shall be strain free, and shall contain no loops or rises which could cause vapor lock. The fuel supply line shall contain an engine manufacturer's standard fuel filter. The fuel line shall be not less than 6 feet long and shall have a quick-disconnect fitting for connecting to the engine fuel inlet. The fuel line shall be gasoline and oil resistant and shall not deteriorate due to aging, sunlight, heat, cold, or salt water. An automatic valve shall be provided in the quick-disconnect fitting to close off the fuel line from the fuel tank when the line is disconnected from the engine.

3.6.5 Electrical system. Engine shall have a generator or alternator to recharge the furnished battery and power all electrical accessories installed on the boat. The ignition system shall be of the manufacturer's standard with protection from contamination and waterproof coils. Ignition points, where applicable, shall be of corrosion-resistant material and shall be adjustable without using special tools. Ignition timing marks shall be provided. All wiring shall be installed to prevent straining or chafing of components during operation. Engine shall be provided with an electric starting mechanism of the manufacturer's standard. Provisions shall be made for manual starting. A heavy-duty marine battery shall be furnished (see 3.5.10).

3.6.6 Exhaust system. The engine manufacturer's standard underwater exhaust system shall be provided.

3.6.7 Cooling system. A positive circulating water-cooling system shall be provided. The system shall be self-draining in the operating position when the outboard motor is not running.

3.6.8 Propeller. The propeller shall have three or four blades and shall be fabricated of manganese bronze, or heat-treated, semipermanent or permanent mold, aluminum-alloy casting, or stainless steel (see 3.6.8.1) as specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

(see 6.2). The size (diameter and pitch) of the propeller shall be selected for compatibility with 3.6.1 and shall be die stamped on the hub. A slip-type clutch, integral with the propeller shall be furnished which will prevent damage to the outboard motor when the propeller strikes an underwater object.

3.6.8.1 Stainless steel propeller. When specified (see 6.2), the propeller shall be fabricated of ASTM A 473, type 316 or ASTM A 743, type CF-3M or CF-8M, stainless steel.

3.7 Trailer design. The utility boat system shall include a commercial boat trailer having a rated capacity of 3,000 lb. The trailer shall be certified by the TMA. The trailer shall include all necessary electrical and mechanical equipment for connection to an available Navy pick-up truck or other towing vehicle. Additionally, all necessary equipment for loading, securing, and unloading the 20-foot utility boat and accessories shall be provided.

3.7.1 Performance. Vehicle performance, as specified in 3.7.1.1 through 3.7.1.5, shall be demonstrated by the trailer when fully equipped with all attachments and accessories and with the 20-foot utility, engine, and accessories loaded aboard. Trailer shall be completely serviced and then towed by a towing vehicle over improved and unimproved hard, uneven terrain. Vehicle shall show no evidence of failure or malfunction of trailer, boat, engine, components, attachment, and accessories.

3.7.1.1 Towing speed and training ability. When transporting the utility boat, engine, and all accessories while being towed over dry, level, smooth, improved, and prepared hard-surface roads at sustained speeds to 50 miles per hour (mph), and over dry unimproved gravel raods at speeds to 10 mph, the trailer shall follow the towing vehicle without swerving from side to side over 3 inches to each side of the path of the towing vehicle when it is moving in a straight line.

3.7.1.2 Manual maneuvering. The trailer shall have provisions for short distance, manual maneuvering by two men.

3.7.1.3 Storage and hookup. The trailer shall have provisions for horizontal storage when not connected to a tow vehicle. When stored fully loaded, the trailer shall be capable of being attached to a trailer hitch equipped pickup truck by one man without requiring any manual lifting.

3.7.1.4 Onloading and offloading. With the trailer attached to the truck, the utility boat with engine and accessories shall be capable of being unfastened from the trailer and offloaded into the water, utilizing the attached electric winch or the manual override feature. Onloading of the boat onto the trailer shall also be performed utilizing either the electric winch or the manual override feature. The trailer bed, tiltdown feature shall be utilized in onloading and offloading. All devices shall perform with no evidence of failure or malfunction.

3.7.1.5 Turning ability. When coupled to a towing vehicle operating in its minimum turning circle, the trailer shall follow without cramping or side

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

slipping. The trailer shall be capable of assuming an angle of 60° with the towing vehicle without damage to either or interference between vehicles.

3.7.2 Trailer components.

3.7.2.1 Suspension. Trailer shall be equipped with a suspension system having a rated capacity at least equal to the load imposed on each member measured at the ground, with vehicle loaded as specified in 3.7. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted.

3.7.2.2 Axle. Axle rating shall be at least equal to the load imposed on the axle measured at the ground with vehicle loaded as specified in 3.7.1.

3.7.2.3 Wheels, rims, tires, and tubes. Vehicle wheels shall be manufacturer's commercial standard. Rims and tire ratings shall conform to T&RA recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each vehicle.

3.7.2.4 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with vehicle loaded as specified in 3.7.1. Tires shall be of not less than 100 level quality. Manufacturer's current size of commercial tires conforming to specified requirements may be furnished.

3.7.2.4.1 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy-duty type and shall be of proper size for tires furnished.

3.7.2.5 Bearing lubrication fittings. External bearing lubrication fittings shall be provided for each hub.

3.7.2.6 Spare wheel assembly. Unless otherwise specified (see 6.2), a spare wheel or rim shall be mounted on a carrier and installed in a readily accessible location. Carrier shall permit ready removal or installation of spare assembly.

3.7.2.6.1 Spare tire. Unless otherwise specified (see 6.2), an inflated spare tire mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size, tread design, and ply rating as the tires in use on the vehicle.

3.7.2.7 Brakes The trailer shall be provided with manufacturer's standard hydraulic surge brakes on each wheel. The brake system shall be provided with an automatic breakaway actuation device.

3.7.2.8 Lighting and electrical systems. Trailer lighting and electrical systems shall comply with the ABYC and NMMA standards.

3.7.2.8.1 Lighting. The electrical lighting system shall be 12-volt and shall conform to ABYC and NMMA standards. Lights and reflectors shall not be mounted on rubrails or vehicle bumpers.

Source: https://assist.dla.mil -- Downloaded: 20260-20T21:47Z
Check the source to verify that this is the current version before use.

3.7.2.8.2 Lighting cable. Trailer lighting connectors, coupling sockets, and cable shall conform to ABYC and NMMA standards. Coupling socket on trailer shall be protected with cover assembly. Lighting cable shall be of sufficient length to connect the trailer to the towing vehicle and allow adequate clearance for turning. Provisions shall be made to secure connector socket and cable to drawbar or cargo body when not in use.

3.7.2.8.3 Turn signals. Turn signal lamps, conforming to ABYC and NMMA standards, shall be furnished.

3.7.2.9 Landing leg. Trailer shall have a retractable landing leg with caster type landing wheel, with cranking handle (rigid or swivel type) permanently installed. The landing leg and related parts shall withstand, without permanent deformation, the combined static and dynamic forces due to the proportionate weight of the gross weight and impacts during coupling and uncoupling operations. Landing leg shall operate freely and shall be protected from malfunctioning by foreign matter. When placed into travel position, the landing leg shall remain positively locked to provide ground clearance in the course of normal towing operations.

3.7.2.10 Coupling device and towbar. Trailer shall be furnished with a ball type coupler. Towbar, coupling device, and safety chains shall be provided and shall conform to ABYC and NMMA standards.

3.7.2.11 Frame. Frame shall be a heavy-duty type of the manufacturer's standard. Frame shall be capable of withstanding all design loadings with no evidence of malfunction or deformation.

3.7.2.12 Fenders. Vehicle shall have fenders over wheels. The fenders shall have inside shields and shall have sufficient crown to control mud and water from splashing on chassis and boat.

3.7.2.13 Trailer bed tilting mechanism. The trailer shall be designed to allow the trailer bed to tilt down while remaining hitched to the truck to allow easier onloading and offloading of the boat in the water. The mechanism to effect this shall be of the manufacturer's standard for the load rating of the trailer.

3.7.2.14 Winch. The trailer shall be equipped with an electric 12-volt dc powered winch with steel cable which shall be rated for loads including 6,000 lb. The winch shall have a manual override which shall be equipped with a hand crank.

3.7.2.15 Hold-down straps. Quick-release nylon web safety straps for holding boat trailer in transit shall be provided.

3.7.3 Towing vehicle equipment. Towing vehicle 2-inch coupling ball and electrical receptacle and wiring harness shall also be provided to allow the system to be connected to an available Navy pick-up truck or other towing vehicle.

3.8 System construction. The system shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories

Source: https://assist.dla.mil -- Downloaded: 20260-20T21:47Z
Check the source to verify that this is the current version before use.

shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.9 Dissimilar metals. Intimate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.10 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations and shall be compatible with lubricants specified herein. Pressure lubrication shall not damage seals or other parts. All parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used.

3.11 Servicing and adjusting. Prior to acceptance of the utility boat system by the Government inspector, contractor shall service and adjust the system components for operational use including at least the following: inflation of all tires; lubrication of all required boat, engine, trailer chassis, and running gear locations with grades of lubricant recommended for the ambient air temperature at the delivery point (see 3.10); adjustment and tuning of engine; adjustment of all cables and controls; and mounting of all components and accessories.

3.12 Corrosion protection.

3.12.1 Trailer structural metal. All trailer structural metal parts shall be fabricated of metal having been protected by corrosion retarding processes other than paint, which are certified as protective in a saltwater neighboring environment.

3.12.2 Metal boat fittings. All metal boat fittings shall be either manganese, bronze, monel, marine aluminum, or stainless steel (300 series) and shall either be inherently corrosion resistant or shall be treated so as to retard corrosion and shall conform to ABYC standards.

3.12.3 Engine metal surfaces. Engine metal surfaces shall be protected from saltwater corrosive environment by the engine manufacturer's standard method of protection.

3.12.4 Fasteners. All metal fasteners which might be exposed to corrosive environments shall be nickel-plated brass or stainless steel (300 series).

3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Interior and exterior fiberglass surfaces of the boat shall be finished using the manufacturer's standard impregnated gel coat. Color of the finish coat for both paint and gel coat shall be light grey unless otherwise specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. For metal

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:47Z
Check the source to verify that this is the current version before use.

surfaces not otherwise protected (see 3.12), painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting, when utilized, shall be with manufacturer's current materials according to manufacturer's current processes and the total dry thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.14 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.15 Marking, identification plates, instruction plates, and boat registry numbers.

3.15.1 Marking. The utility boat system shall be marked in accordance with MIL-STD-130 unless otherwise required by the contracting officer (see 6.2).

3.15.2 Identification plates. An identification plate will be furnished by the contracting officer for each utility boat system. Nomenclature shall be UTILITY BOAT SYSTEM, (component name). The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose and securely affix a plate to each utility boat system in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8 inch in diameter. A plate shall be included for the trailer, boat, and outboard engine components of each system.

3.15.3 Instruction plates. The utility boat system components shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.15.4 Boat registry numbers. Boat registry numbers to be assigned after contract award, when required, shall be 3 inches high, black painted numbers. The numbers shall be located as required to obtain the required certificate of numbers from the Coast Guard or the State of Registration.

3.16 Workmanship. All parts, components, and assemblies of the utility boat system including castings, forgings, molded parts, stamping, bearings, seals, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, and other harmful extraneous material. External surfaces shall be free from burrs, dents, chipped paint, and rough or sharp edges and corners. Fasteners shall be tight without distorting the components being held. Operable components shall function without binding or causing interference of movement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.16.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in areas of shear and vibration, and all bolts, nuts and screws shall be tight.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, nearly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Welding. Welding procedures shall be in accordance with AWS D1.1 or D1.2, as appropriate. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article inspection shall be performed on one Utility Boat System when a first article is required (see 3.2 and 6.2). This first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.4 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the tests of 4.6, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.3.

4.5 Examination. Each utility boat system shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified maximum efficiency shall constitute cause for rejection.

4.6 Tests.

4.6.1 General. A compliance check for presence of all accessories specified and compatibility with stated required certifications shall be made.

4.6.2 Floor plan. A check for compliance with specified interior floor plan configuration shall be made.

4.6.3 Mechanisms and accessories. All mechanisms, mechanical and electrical, shall be operated to determine acceptability of smooth operation. Accessories mounted on brackets shall be removed and relocated on the brackets to determine acceptability of mounting configuration.

4.6.4 Hoisting. The boat and engine, while loaded with subject accessories and fully filled fuel tanks, shall be mechanically hoisted into the water, off the trailer, from a pier at least 10 feet above the water surface. Hoisting fittings and the mounting areas immediately adjacent to the fittings

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

shall be visually inspected, after boat has been launched and retrieved, for signs of strain of hardware cracking or other damage to hull. Damage to these items shall constitute grounds for failure of the test.

4.6.5 Launching and retrieving. The boat and engine, while fully loaded with accessories and filled fuel tanks, shall be launched from the trailer when driven onto a launching ramp. The equipped boat shall then be retrieved onto the trailer using the electric powered winch. The boat shall then be retested using the hand override winch system. In both cases, the trailer bed tiltdown mechanism shall be utilized. All launching and retrieving equipment shall perform easily and speedily. Permanent deformation, excessive binding, (or damage to any portion of the system) will constitute failure of this test. This test will determine operational acceptability of the trailer launching assembly.

4.6.6 Towing device. The contractor shall demonstrate to the Government representatives the capacity of the towing device to sustain six times the rated thrust of the boat-engine-propeller combination for 5 minutes without causing visible damage to the hull structure in the areas around the connections to the hull or to any other area of the towing device. Device shall be inspected to ensure tow ropes do not interfere with engine.

4.6.7 Road test. The trailer with rated payload shall be road tested for a distance of not less than 10 miles to determine conformance to the requirements of this specification.

4.6.8 Storage, hookup, and manual maneuvering. Horizontal storage, short distance manual maneuvering, and hookup shall be demonstrated to determine conformance to the requirements of this specification. Hookup shall include electrical and mechanical coupling of the trailer and towing vehicle.

4.6.9 Operating test. The boat shall be deployed and in-water compliance checked for mechanism, control, gauge, and engine operation. Included in this test shall be seaworthiness, handling of the boat, and a static thrust test of the engine propeller combination.

4.7 Packaging inspection. The preservation, packing, and marking of the boat system shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation and packaging. The preservation and packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to protect parts known to be subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, screws, pins, and washers shall be reinstalled in mating parts and secured to prevent their loss.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1.

5.1.1.4 Outboard motor. With the engine running smoothly at approximately 1,500 rpm, P-10 preservative shall be sprayed into the carburetor throat(s) until the engine chokes to a stop. A tag conforming to UU-T-81, type B, class 2 shall be attached to a conspicuous location indicating that "ENGINE PRESERVATION; DO NOT CRANK OR RUN." The carburetors and cooling system shall be drained dry.

5.1.1.5 Fuel tanks. The fuel tanks shall be drained of all fuel and dried. The interior of the tanks shall be coated with type P-10 preservative in a manner to ensure thorough coating of all surfaces. The tanks shall be secured to the floor with the straps provided. The detached fuel lines shall be packed in a fiberboard box conforming to PPP-B-636, class weather resistant and cushioned to prevent movement.

5.1.1.6 Battery. The battery shall be placed inside the battery case and cushioned to prevent movement. The battery case with cover installed shall be secured to the boat floor with the straps provided.

5.1.1.7 Life preservers. The life preservers shall be packed in fiberboard boxes conforming to PPP-B-636, class weather resistant and cushioned to prevent movement.

5.1.1.8 Wood paddles, boat hook, fenders, and anchor. The paddles, boat hook, fenders, and anchor shall be secured in a protected location inside the boat in a manner to prevent movement and damage during shipment.

5.1.1.9 Winches. Unreel and coat all surfaces so that the rope is completely immersed in type P-1 preservative. Damaged preservative coating resulting from rewind shall have additional preservative applied.

5.1.1.10 Rope. The rope shall be coiled or hanked and secured with cotton cord or twine.

5.1.1.11 Fire extinguisher. The fire extinguisher shall be placed and secured in the mounting brackets.

5.1.1.12 First-aid kit. The first-aid kit shall be preserved in accordance with MIL-P-116 method IC-2.

5.1.1.13 Emergency flare kit. The kit shall be preserved in accordance with MIL-P-116, method III, in a fiberboard box conforming to PPP-B-636, class weather resistant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

5.1.1.14 Spotlights and running lights. Spotlights and running lights, if removed, shall be preserved as specified for the emergency flare kit in 5.1.1.13.

5.1.1.15 Repair parts. The preservation application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.1.16 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packing requirements of PPP-P-40.

5.1.1.17 Consolidation.

5.1.1.17.1 Repair parts, maintenance tools, and small components. Repair parts, tools, and small components shall be packaged in close-fitting boxes conforming to PPP-B-601, domestic type or PPP-B-636, class weather resistant. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

5.1.1.17.2 Outboard Motor. The outboard motor shall be packed in a close-fitting box conforming to PPP-B-601, domestic type. The motor shall be cushioned, blocked, and braced to prevent movement within the box.

5.1.2 Commercial. The complete boat system shall be preserved in a manner that will ensure protection against deterioration and damage during shipment. This level may conform to the contractor's standard practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The safety equipment, disassembled components, and consolidated packaged components shall be placed inside the boat in a protected location, and blocked and braced to prevent movement during shipment and damage to the boat. Components in wooden boxes shall be secured to the base of the crate or placed and secured on the deck of the boat. The canvas cover shall be installed in place in a manner to prevent water pockets. The boat shall be placed on the trailer, secured with the means provided, and packed in a crate conforming to MIL-C-104, type II, class optional, style A. Cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186 and the appendix to the crate specification.

5.2.2 Level B. Each complete boat system shall be packed as specified for level A, except the crate shall conform to MIL-C-3774, type II.

5.2.3 Commercial. The complete boat system shall be packed for shipment in a manner which will ensure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The utility boat system is intended to be used to tow oil spill booms into place. The system is designed to be towed to the site by an available towing vehicle and the boat launched either from the trailer on a boat launching ramp or by the dock crane.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required for inspection approval (see 3.2, 4.3 and 6.3).
d. When outboard mounting well shroud is required (see 3.5).
e. When the console shall be positioned other than on the starboard side of the boat (see 3.5.4.1).
f. When water temperature indication is required (see 3.5.4.1.3).
g. When ammeter is required (see 3.5.4.1.4).
h. When engine log meter is required (see 3.5.4.1.5).
i. Whether propeller shall have 3 or 4 blades (see 3.6.8).
j. Whether propeller shall be fabricated of manganese bronze or heat-treated semipermanent or permanent mold aluminum-alloy casting or stainless steel (see 3.6.8 and 3.6.8.1).
k. When spare wheel assembly is not required (see 3.7.2.6).
l. When spare tire is not required (see 3.7.2.6.1).
m. When finish color shall be other than light grey (see 3.13).
n. When fungus resistance is required (see 3.14).
o. When marking other than in accordance with MIL-STD-130 is required (see 3.15.1).
p. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample or a standard production item from the contractor's current inventory as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:47Z
Check the source to verify that this is the current version before use.

6.4 Subject term (keyword) listing.

Boat
Outboard
Vessel

6.5 Changes from the previous issue. Marginal notations not used in this revision to identify changes with respect to the previous issue to the extensiveness of the changes.

Preparing activity
Navy - YD

(Project No. 1940-N073)

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-29159A(YD)	MILITARY SPECIFICATION _ BOAT SYSTEM, UTILITY, NOMINAL, 20-FOOT REINFORCED PLASTIC WITH 100-HORSEPOWER OUTBOARD ENGINE AND TRAILER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 20260-20T21:47Z
Check the source to verify that this is the current version before use.